

Data Storytelling

Turn Numbers Into Narratives That Drive Decisions

■ COMMUNICATION & NARRATIVE

■ LEARNING OUTCOME

By the end of this PDF, you will structure compelling data stories using proven frameworks, design visuals that communicate clearly, and present insights to any audience — technical or not.

■ Skills You Will Gain

- Use the SCR & Pyramid frameworks for storytelling
- Design charts that make one clear point
- Apply the 5-second rule to every visual
- Present to non-technical stakeholders
- Write insight statements that lead with the 'so what'
- Build executive-ready slide decks

■ Analysts who communicate well earn 2x more. Most analysts can find insights. Very few can explain them. That gap is your advantage.

SECTION 1

What Is Data Storytelling?

Data Storytelling is the skill of combining **data + visuals + narrative** to communicate insights in a way that inspires action. Raw numbers don't move people — stories do.

Component	Role in the Story
Data	The evidence — makes your claim credible and verifiable
Visuals	The hook — makes patterns visible and memorable
Narrative	The meaning — explains WHY it matters and WHAT to do

SECTION 2

The SCR Framework

1	Situation (S) Set the context everyone already knows. Example: 'Our Q3 e-commerce revenue was ■45L, same goal as Q2.'
2	Complication (C) Introduce the tension — what changed or went wrong. Example: 'But cart abandonment jumped from 60% to 78%, meaning we lost ■15L in potential revenue.'
3	Resolution (R) Deliver the insight + recommendation clearly. Example: 'Fixing the checkout flow UX can recover ■12L — we need a 2-week sprint approved now.'

Why SCR Works

The human brain is wired for story structure: setup → conflict → resolution. SCR mirrors this pattern, so your audience stays engaged and remembers the message.

SECTION 3

The Pyramid Principle

Always lead with your **conclusion first**, then support it with evidence. Never bury the insight at the end.

1	Answer First Start with your main recommendation: 'We should expand to Tier-2 cities.'
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2	Key Arguments Give 3 supporting points: market size, low competition, rising income
3	Data Evidence Each point backed by specific data: 'Tier-2 cities grew 34% YoY in app installs'

Wrong approach (bottom-up): 'We looked at 12 months of data... ran 3 analyses... considered 5 scenarios... and therefore we recommend Tier-2 expansion.' **Right approach (top-down):** 'We recommend Tier-2 expansion. Here's why: [3 bullets]. Here's the data.'

SECTION 4

Visualization Best Practices

Rule	Application
One message per chart	Ask: 'What is the ONE thing this chart should show?' Remove everything else.
Start Y-axis at zero	Truncating axes makes tiny changes look massive — misleads the audience.
Label directly	Put labels on the bars/lines, not in a legend that forces eye-movement.
Color with purpose	Red = problem, Green = good, Blue = neutral. Never use 7 colors.
Remove chartjunk	Delete unnecessary gridlines, 3D effects, background images, borders.
Add the insight	Put a title like 'North Revenue Grew 34% — Highest Region' not just 'Revenue by Region'.
The 5-Second Rule	If the main message isn't clear in 5 seconds, simplify the visual.

SECTION 5

Choosing the Right Chart

What You Want to Show	Best Chart Type
Comparison between categories	Bar Chart / Column Chart
Change over time / trend	Line Chart
Part of a whole (proportions)	Pie Chart (max 5 slices) or Stacked Bar
Relationship between two variables	Scatter Plot
Distribution of values	Histogram or Box Plot
Progress toward a goal	Gauge Chart or KPI Card
Geographic patterns	Choropleth Map or Bubble Map
Ranking or ordered comparison	Horizontal Bar Chart (sorted)

SECTION 6

Writing Insight Statements

Weak insight (describes the data):

"North region had sales of █15L and South region had █9L."

Strong insight (explains meaning + implication):

"North region outperformed South by 67% — driven by 3 high-value enterprise clients. Replicating this client-acquisition approach in South could add █6L by Q4."

The 3-Part Insight Formula

1. WHAT: State the observation with a number 2. SO WHAT: Explain why it matters to the business 3. NOW WHAT: Give one clear recommended action

SECTION 7

Presenting to Non-Technical Audiences

1	Replace jargon 'p-value = 0.03' → 'We are 97% confident this change is real, not luck'
2	Lead with impact Don't say 'We ran a regression.' Say 'Revenue will drop 12% next quarter.'
3	Use comparisons 'Revenue is █12L' is weak. '█12L — up 35% vs last year' is strong.
4	One slide, one point Executive presentations: 1 insight per slide, max 5 slides.
5	End with action Close every presentation: 'The one thing we need you to approve is...'

SECTION 8

The Storytelling Slide Structure

Slide	Content
Slide 1: Hook	The business problem or opportunity in one sentence + striking number
Slide 2: Evidence	Your key chart or data — the visual proof of the problem/opportunity
Slide 3: Root Cause	'Why is this happening?' — supporting analysis
Slide 4: Recommendation	Your proposed solution with expected impact (ROI, timeline)
Slide 5: Next Steps	3 specific actions, owner name, and deadline for each

█ PRACTICE QUESTIONS

Q1. What are the 3 components of data storytelling?

Answer: Data (evidence), Visuals (hook), and Narrative (meaning and action).

Q2. In the SCR framework, what does 'Complication' mean?

Answer: The tension or problem that disrupts the normal situation — the 'but' that makes people pay attention.

Q3. Why should you never truncate the Y-axis on a bar chart?

Answer: It makes small differences appear dramatically large, misleading the audience about the actual magnitude of change.

Q4. Rewrite this weak title: 'Sales by Region Q3'

Answer: 'North Region Leads With █15L — 67% Above South' — always put the insight in the title.

Q5. What is the Pyramid Principle?

Answer: Lead with your conclusion/recommendation first, then support it with arguments and data. Never bury the insight at the end.

■ BONUS RESOURCES

Book	'Storytelling with Data' by Cole Nussbaumer Knaflc — the #1 book on this topic
Book	'The Pyramid Principle' by Barbara Minto — structure for business communication
Tool	Canva / PowerPoint — build clean executive presentations
Practice	Take any chart you see and rewrite its title as an insight statement
YouTube	Search 'Storytelling with Data' channel — free walkthroughs
Template	Use the SCR framework for every insight you share at work or in interviews

■ Data Storytelling Cheat Sheet

SCR Framework	Situation → Complication → Resolution
Pyramid Principle	Conclusion first, evidence second
5-Second Rule	Message clear in 5 seconds or simplify
Insight Formula	WHAT + SO WHAT + NOW WHAT
Trend over time	Use Line Chart
Compare categories	Use Bar/Column Chart
Part of whole	Use Pie (max 5 slices)
Correlation	Use Scatter Plot
Distribution	Use Histogram/Box Plot
Color: Red	Problem / negative / alert
Color: Green	Good / positive / on track
Remove	Gridlines, 3D, legends when possible